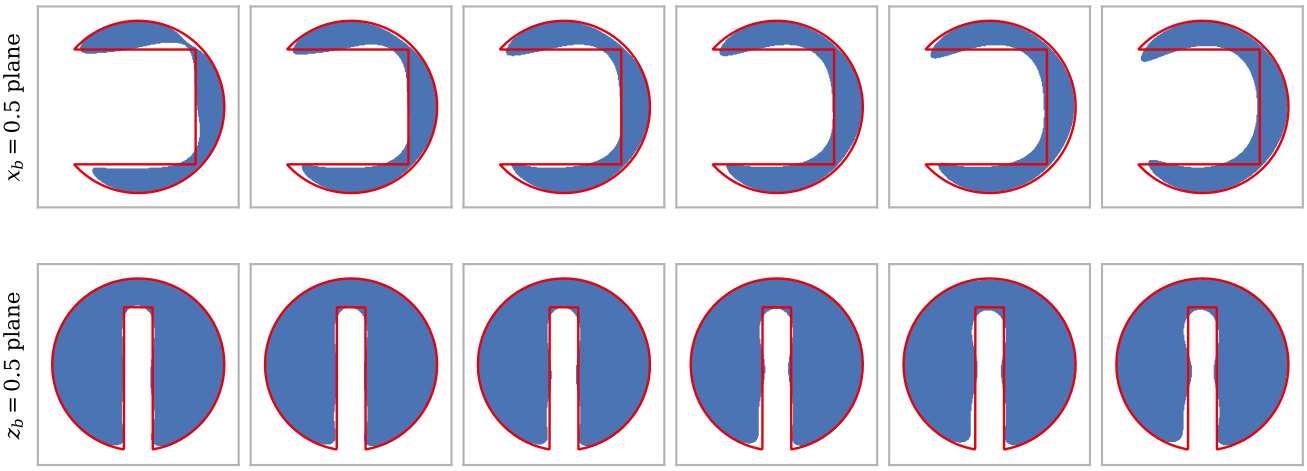

$t = 0$
$t = T/5$
$t = 2T/5$
$t = 3T/5$
$t = 4T/5$
$t = T$
$x_b = 0.5$ plane
$z_b = 0.5$ plane
PINN solid $\hat{\phi} < 0$
exact $\phi = 0$